Lab 2.1 Technical specification with RISE

IES | 20 minutes | LO2, LO3

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

Inputs and preparation

Open the mock-data folder: design-requirements.csv, DES-01-rev2.md, claim-register.csv. Use prompts.md for copy and paste. Save your reviewed work under outputs. Earlier lab outputs are optional; this folder provides standalone inputs.

Procedure

1. Open design-requirements.csv and DES-01-rev2.md. Review confirmed requirements versus gaps. Create a Word document in approved storage named Lab-2.1-Specification.

2. In the Copilot pane attach or reference the approved source files if available; otherwise paste their short text. State the source names explicitly. Use the R.I.S.E. prompt below.

3. Review the draft section by section. Check nominal flow 50 m3/h, head 30 m and the teaching tolerance. Keep wetted material, supply voltage and ambient conditions as TBD.

Lab 2.1 continued

4. Ask Copilot to produce a requirement-to-test checklist with IDs, values, units, source clause, evidence and status. Independently compute 50 x 0.95 and 50 x 1.05; compare with the acceptance band.

5. Run the verification prompt in prompts.md. Correct any invented material or standard. Save the specification and checklist with a draft status and reviewer field.

Check before moving on

Teaching band is 47.5-52.5 m3/h.

Every requirement cites its source or is marked missing.

No invented material, voltage or approval.

Access or source problems

Use approved paired access or the local source files. Record unavailable live capabilities as not executed. If an attachment is not supported, paste the short synthetic text or reference its approved SharePoint copy. Do not use personal accounts to bypass controls.

Lab 2.1 prompts and evidence

Main prompt

Role: engineering specification writer. Input: design-requirements.csv and DES-01 rev2 only. Steps: extract confirmed requirements; separate gaps; draft scope, duty, materials, acceptance and review sections; create a requirement-to-test checklist. Expectations: preserve IDs and units, cite clauses, leave missing fields TBD, and mark DRAFT FOR REVIEW. DES-01 is a fictional design note, not a published standard.

Verification prompt

Audit the draft against the named source files. List each claim, source location, uncertainty and correction. Recalculate key values independently. Return a revised draft and change log; do not approve the engineering deliverable.

Expected output example

Illustrative teaching example, not a recorded AI run. R01: 50 m3/h at the stated duty; source DES-01 rev2 clause 2. R03: 47.5-52.5 m3/h teaching band; record three readings, instrument and witness. R04: material TBD pending compatibility review.

Retain your evidence

Keep the source versions, prompt, tool, output, corrections, independent checks and reviewer decision. Record any capability or permission test that was not executed. Use templates/prompt-log.csv. Do not publish real workplace data.